Lab 1 Bash Scripting

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Using sed utility

1:

```
mohamed@Mohamed-Sameh-PC:~$ sed -n 'lp/p' /etc/passwd
sed: -e expression #1, char 2: extra characters after command
mohamed@Mohamed-Sameh-PC:~$ sed -n '/lp/p' /etc/passwd
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
mohamed@Mohamed-Sameh-PC:~$
```

2:

```
mohamed@Mohamed-Sameh-PC:~$ sed '1,3d' /etc/passwd
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:116::/run/uuidd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:118::/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124::/nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:120:126:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
```

3:

```
Activities  Terminal                Feb 22  12:31 PM                en  68 %
mohamed@Mohamed-Sameh-PC: ~
mohamed@Mohamed-Sameh-PC:~$ sed '$d' /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:116::/run/uuidd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:118::/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124::/nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
```

4:

```
Activities  Terminal                Feb 22  12:36 PM                en  63
mohamed@Mohamed-Sameh-PC: ~
mohamed@Mohamed-Sameh-PC:~$ sed '/lp/d' /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
```

5:

```
mohamed@Mohamed-Sameh-PC:~$ sed 's/lp/mylp/g' /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
mylp:x:7:7:mylp:/var/spool/mylpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
```

Using awk utility

1:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{print $5}' /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
Mailing List Manager
ircd
Gnats Bug-Reporting System (admin)
nobody
systemd Network Management,,,
systemd Resolver,,,
```

2:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{print NR, $1, $5, $6}' /etc/passwd
1 root root /root
2 daemon daemon /usr/sbin
3 bin bin /bin
4 sys sys /dev
5 sync sync /bin
6 games games /usr/games
7 man man /var/cache/man
8 lp lp /var/spool/lpd
9 mail mail /var/mail
10 news news /var/spool/news
11 uucp uucp /var/spool/uucp
12 proxy proxy /bin
13 www-data www-data /var/www
14 backup backup /var/backups
15 list Mailing List Manager /var/list
16 irc ircd /run/ircd
17 gnats Gnats Bug-Reporting System (admin) /var/lib/gnats
18 nobody nobody /nonexistent
19 systemd-network systemd Network Management,,, /run/systemd
20 systemd-resolve systemd Resolver,,, /run/systemd
```

3:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{if ($3 > 500) print NR, $1, $3, $5}' /etc/passwd
18 nobody 65534 nobody
48 mohamed 1000 Mohamed Sameh,,,
mohamed@Mohamed-Sameh-PC:~$
```

4:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{if ($3 == 500) print NR, $1, $3, $5}' /etc/passwd
mohamed@Mohamed-Sameh-PC:~$
```

5:

```
mohamed@Mohamed-Sameh-PC:~$ awk '{for (i=5 ; i<=15; i++) if(NR==i) print $0}' /etc/passwd
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
mohamed@Mohamed-Sameh-PC:~$
```

6:

```
mohamed@Mohamed-Sameh-PC:~$ awk '{for (i=5 ; i<=15; i++) if($i~/lp/) $i=="mylp"}' /etc/passwd
mohamed@Mohamed-Sameh-PC:~$
```

7:

```
mohamed@Mohamed-Sameh-PC:~$ awk '{if ($3 > max) {max = $3; info = $0;}} END {print info}' /etc/passwd
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
mohamed@Mohamed-Sameh-PC:~$
```

8:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{sum += $3} END {print sum}' /etc/passwd
70352
mohamed@Mohamed-Sameh-PC:~$
```

Bonus

1:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{sum[$4] += $3} END {for (group in sum) print "Group ID: " group, "Sum: " sum[group]}' /etc/passwd
Group ID: 121 Sum: 114
Group ID: 102 Sum: 100
Group ID: 34 Sum: 34
Group ID: 3 Sum: 3
Group ID: 1000 Sum: 1000
Group ID: 29 Sum: 119
Group ID: 10 Sum: 10
Group ID: 8 Sum: 8
Group ID: 122 Sum: 115
Group ID: 139 Sum: 129
Group ID: 125 Sum: 118
Group ID: 38 Sum: 38
Group ID: 46 Sum: 111
Group ID: 111 Sum: 104
Group ID: 126 Sum: 120
Group ID: 103 Sum: 101
Group ID: 2 Sum: 2
Group ID: 41 Sum: 41
```

2:

```
mohamed@Mohamed-Sameh-PC:~$ awk -F: '{user[$4]=user[$4]?user[$4]"\n"$1:$1} END{print "User-Group Report:\n";for(group in user)print "GID " group ":\n
 user[group]"\n"}' /etc/passwd
User-Group Report:

GID 121:
avahi

GID 102:
systemd-network

GID 34:
backup

GID 3:
sys

GID 1000:
mohamed
```